FIFTH EDITION

A GUIDE TO PERSONNEL QUALIFICATION & CERTIFICATION

The American Society for Nondestructive Testing

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A **GUIDE** TO PERSONNEL QUALIFICATION & CERTIFICATION

The American Society for
Nondestructive Testing Inc.

ACKNOWLEDGMENTS

The American Society for Nondestructive Testing Inc. is grateful for the volunteer contributions, technical expertise, knowledge, and dedication of the following individuals who have helped make this work possible.

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First printing 1/23
ebook 1/23

Errata, if available for this printing, may be obtained from ASNT's website, asnt.org. Ebooks contain all corrections and updates, including the latest errata.

ISBN: 978-1-57117-589-2 (print)
ISBN: 978-1-57117-590-8 (ebook)

Printed in the United States of America

Published by:
The American Society for Nondestructive Testing Inc.
1201 Dublin Road
Suite #G04
Columbus, OH 43215
asnt.org

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ASNT Mission Statement:
ASNT's mission is to advance the field of nondestructive testing.

ASNT *Code of Ethics*:
The ASNT *Code of Ethics* was developed to provide members of the Society with broad ethical statements to guide their professional lives. In spirit and in word, each ASNT member is responsible for knowing and adhering to the values and standards set forth in the Society's *Code*. More information, as well as the complete version of the *Code of Ethics*, can be found on ASNT's website, asnt.org.

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A GUIDE TO PERSONNEL QUALIFICATION & CERTIFICATION

Introduction

The world of nondestructive testing (NDT) is complex. We utilize highly scientific processes to nondestructively evaluate a variety of materials and components for virtually every industry.

Because of the complex nature of NDT and the vast variety of testing methods at our disposal, it quickly becomes a daunting task to ensure technicians are both highly competent and highly capable. It is paramount that NDT technicians are proficient at performing complicated tests both accurately and consistently.

This results in qualification and certification schemes that can be equally complex. This guide was developed by NDT experts to help assist those seeking to better navigate the complexities of NDT qualification and certification.

Within this guide we will review the most common certification and qualification means as developed by the American Society for Nondestructive Testing (ASNT) as well as review international certification schemes and various specific industries' certification programs.

Definitions

As with most specialized occupations, NDT has its own set of terms, definitions, and acronyms that the reader should be familiar with to properly understand the qualification and certification processes. The definitions below were extracted from several different reference documents which are listed at the end of this section.

- **Basic Examination[1]:** Written examination, at Level III, which demonstrates the candidate's knowledge of the materials science and process technology and types of discontinuities, the specific qualification and certification system, and the basic principles of NDT methods as required for Level II.
- **Certification[2]:** Written testimony that an individual has met the applicable requirements of this standard.
- **Certification Procedure[2]:** A written procedure, developed by the employer, that details the requirements for qualification and certification of an employee to a national standard.
- **Certification Standard:** A document that provides *requirements* for the establishment of a qualification and certification program. The requirements may not be modified but must be followed as written.
- **Education[2]:** An institutionalized program, prescribed by appropriate authorities, that is offered by schools, institutes, organizations, colleges, or universities established for the sole purpose of providing instruction in an orderly, planned, and systematic fashion.
- **Employer[2]:** The corporate, private, or public entity, which employs personnel directly or indirectly for wages, salary, fees, or other considerations. This would include organizations which obtain their qualified supplemental workforce personnel through third-party agencies, provided the use and certification of those supplemental employees is addressed in the employer's certification procedure.
- **Examination[2]:** A formal, controlled, documented assessment of knowledge or skills conducted in accordance with a procedure.
- **Experience[2]:** Actual performance of an NDT method conducted in the work environment resulting in the acquisition of knowledge and skill. This does not include formal classroom training but may include laboratory and on-the-job training (OJT) as defined by the employer's certification procedure.
- **Formal Training[2]:** An organized and documented program of activities designed to impart the knowledge and skills required to be qualified to this standard. Formal training may be a mix of classroom, practical, and programmed self-instruction as approved by the responsible NDT Level III.
- **General Examination[2]:** A written examination addressing the basic principles of the applicable NDT method.
- **Method[2]:** One of the disciplines of NDT; for example, ultrasonic testing, within which various test techniques may exist.
- **Method Examination[1]:** Written examination, at Level III, which demonstrates the candidate's general and specific knowledge, and the ability to write procedures for the NDT method as applied in the industrial or product sector(s) for which certification is sought.
- NDT, NDE, NDI: These initialisms stand for nondestructive testing, nondestructive examination, and nondestructive inspection, respectively. Usage varies among industries, but all relate to the same processes, and all have the same meaning. For simplicity, NDT will be used throughout this guide.

- **Nondestructive Testing[3]:** A process that involves the inspection, testing, or evaluation of materials, components, and assemblies for materials' discontinuities, properties, and machine problems without further impairing or destroying the part's serviceability. Throughout this document the term NDT applies equally to the NDT inspection methods used for material inspection, flaw detection, or predictive maintenance (PdM) applications.
- **Practical Examination[2]:** An examination used to demonstrate an individual's ability in conducting the NDT methods that will be performed for the employer. For Practical examinations, questions and answers need not necessarily be written, but observations and results must be documented.
- **Qualification[2]:** The education, skills, training, knowledge, and experience required for personnel to properly perform to a specified NDT level.
- **Recommended Practice[3]:** A set of guidelines to assist the employer in developing uniform procedures for the qualification and certification of NDT personnel to satisfy the employer's specific requirements.
- **Specific Examination[2]:** A written examination to determine an individual's understanding of procedures, codes, standards, specifications, and equipment or instrumentation for an NDT method used by the employer.
- **Technique[2]:** A category within an NDT method; for example, immersion ultrasonic testing.
- **Written Practice[3]:** A written procedure developed by the employer that details the requirements for qualification and certification of its employees.

REFERENCES

1. ISO, 2012, ISO 9712:2012 *Non-destructive testing—Qualification and certification of NDT personnel*, International Organization for Standardization, Geneva, Switzerland
2. ASNT, 2020, ANSI/ASNT CP-189: *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel*, American Society for Nondestructive Testing, Columbus, OH
3. ASNT, 2020, Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing*, American Society for Nondestructive Testing, Columbus, OH

Other common references that contain common NDT definitions

ASTM, latest edition, ASTM E1316 – *Standard Terminology for Nondestructive Examinations,* Section A – *Common NDT Terms*, ASTM International, West Conshohocken, PA

ASME, latest edition, *ASME Boiler and Pressure Vessel Code*, Section V: *Nondestructive Examination*, Article 1, *Mandatory Appendix I: Glossary of Terms for Nondestructive Examination*, American Society for Mechanical Engineers, New York, NY

ISO, 2019, ISO/TS 25107:2019 *Non-destructive testing—NDT training syllabuses*, International Organization for Standardization, Geneva, Switzerland

The Process of Personnel Qualification and Certification

As will be discovered, qualification and certification as an NDT technician can be rather complicated. The document or organization governing certification will affect the various requirements necessary to meet Level I, II, or III status.

Luckily, the vast majority of governing documents/organizations outline the same general path to achieving various NDT levels. All certification schemes will require a visual acuity and color/grayscale examination, though the specific examinations may vary.

- **Education/Training:** It is necessary to obtain a certain number of training hours in each NDT method for each level. The number of training hours will vary for each NDT method and those hours may vary depending on the certification scheme, but all require a specific amount of training. Training topical outlines should be written to a recognized standard such as ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel*.
  Note: ASNT Recommended Practice No. SNT-TC-1A: *Personnel Qualification and Certification in Nondestructive Testing* does not require additional training for Level III qualification; however, ISO 9712: 2012 *Non-destructive testing—Qualification and certification of NDT personnel* does.
- **Experience:** Each NDT level requires a specific amount of experience for each method. Again, this experience will vary for each certification scheme, but all require some level of experience. This experience is typically OJT where the individuals work alongside a certified technician (Level II or III) gaining knowledge and practice in real-world situations.
- **Examination:** Before becoming certified, each prospective NDT technician is required to take a series of exams, typically a General, Specific, and Practical. The number of questions for these exams will vary depending on the certification scheme and method.

Education/training, experience, and examination comprise the fundamental base to all NDT programs and every standard/organization follows this format with some slight variations. Most schemes do not specify the completion order for the training, experience, and examinations. One common sequence is for a trainee to receive their initial training in a method first, then take the General, Specific, and Practical examinations. Assuming all examinations are passed, the trainee will work with a certified technician until the minimum experience hours are met. Another option is for an individual to gain some or all supervised (by a Level II or Level III) experience hours prior to structured training and examinations.

Typical NDT Levels

Prior to certification to any level, a visual examination typically consisting of near-vision acuity and color deficiency or grayscale recognition is required.

- **Trainee:** A trainee is an individual in the process of being initially trained, qualified, and certified and is not yet certified to any level. A trainee works with a certified individual to gain the necessary instruction while accruing the necessary experience hours. The trainee does not independently conduct, interpret, evaluate, or report the results of any NDT test. The trainee is the first step in the NDT certification process.
- **NDT Level I:** An NDT Level I is an individual who has progressed through the trainee stage in any given NDT method. The NDT Level I has gained the necessary classroom training and on-the-job experience to meet the employer's requirements for NDT Level I certification in the specific method and technique. The NDT Level I must pass a General, Specific, and Practical examination to become certified. Once certified, the NDT Level I individual has sufficient technical knowledge and skills to be qualified to properly perform specific standardizations and specific tests. With written approval of the NDT Level III, the NDT Level I may perform specific evaluations for acceptance or rejection determinations according to written instructions and to record results. The NDT Level I is able to follow approved NDT procedures and must receive the necessary instruction and supervision from a certified NDT Level II or III individual.
- **NDT Level II:** An NDT Level II is an individual who has progressed through the trainee and NDT Level I stages in any given NDT method. In some instances, the individual may progress directly from trainee to NDT Level II provided the combined requirements are met. The NDT Level II must complete the necessary classroom training and on-the-job experience to meet the employer's requirements for NDT Level II certification in the specific method and technique. The NDT Level II must pass a General, Specific, and Practical examination to become certified. Once certified, the NDT Level II has the skills and knowledge to set up and standardize equipment; to conduct tests; and to interpret, evaluate, and document results in accordance to applicable codes, standards, and specifications. The NDT Level II is thoroughly familiar with the scope and limitations of the methods and techniques to which certified and is capable of directing the work of trainees and NDT Level I personnel. The NDT Level II organizes and reports nondestructive test results.
- **NDT Level III:** An NDT Level III is an individual who has progressed through the NDT Level II stage in any given NDT method. The NDT Level III must have met the employer's requirements for experience beyond the NDT Level II requirements in NDT in an assignment comparable to that of an NDT Level II in the applicable method. The NDT Level III individual has sufficient technical knowledge and skills to be capable of developing, qualifying, and approving procedures; establishing and approving techniques; interpreting codes, standards, specifications, and procedures; and designating the particular NDT methods, techniques, and procedures to be used. The NDT Level III is responsible for the NDT operations for which qualified and assigned and is capable of interpreting and evaluating results in terms of existing codes, standards, and specifications. The NDT Level III has sufficient practical background in applicable materials, fabrication, and product technology to establish techniques and to assist in establishing acceptance criteria when none are otherwise available. The NDT Level III has general familiarity with other appropriate NDT methods, as demonstrated by an ASNT Level III Basic examination or Basic examination administered by the employer. The NDT Level III, in the methods in which certified, has sufficient technical knowledge and skills to be capable of training and examining NDT Level I, II, and III personnel for certification in those methods. In some instances, the NDT Level III is labeled as the employer's certifying authority and has ultimate responsibility for the written practice and the certification of all employees.
- **Limited Level Certification:** It is common practice in many industries to certify individuals to a limited level. An example is NAS 410: *National Aerospace Standard Certification & Qualification of Nondestructive Test Personnel*, which permits an NDT Level I Limited, which is a limited certification allowing only the performance of a specific NDT test on a specified part, part feature, or assembly. Additionally, the certification of an NDT Level II Limited is quite common; this is an individual who has progressed through the trainee and NDT Level I stages in any given NDT method. In some instances, the individual may progress directly from trainee to NDT Level II Limited. The NDT Level II Limited has gained the necessary classroom training and on-the-job experience to meet the employer's requirements for NDT Level II Limited certification in the specific method and technique. The NDT Level II Limited must pass a General, Specific, and Practical examination plus a vision examination to become certified. Once certified, the NDT Level II Limited has the skills and knowledge to set up and standardize equipment, to conduct only the specific tests detailed in the limited certification, and to interpret, evaluate, and document results in accordance with procedures approved by an NDT Level III. The NDT Level II Limited is thoroughly familiar with the scope and limitations of the technique to which certified and is capable of directing the work of trainees and NDT Level I personnel. The NDT Level II Limited organizes and reports nondestructive test results within the limitations of the certification. There are also some industries that recognize an NDT Limited Level III, though this is rare. The majority of NDT certification documents allow limited level certifications.

- **NDT Instructor:** The NDT instructor has the skills and knowledge to plan, organize, and present classroom, laboratory, demonstration, and/or OJT NDT instruction, training, and/or education programs in accordance with course outlines approved by an NDT Level III.

The Importance of Personnel Qualification and Certification

It is paramount that NDT personnel understand the seriousness of their activities and the consequences of not performing the tests properly. Personal integrity is an absolute necessity, and to help ensure that NDT personnel understand the seriousness of their actions, ASNT requires all personnel applying to take ASNT examinations to agree to abide by a code of ethics.

The qualification/certification process and examinations described in this document are important because a properly administered qualification and certification program demonstrates to NDT service users that the organization's certified individuals thoroughly understand the technology, principles, and practices of the NDT method being used, as well as the proper use of testing techniques. The ability to properly perform NDT and to accurately report the results is essential because

- many NDT methods do not inherently create their own records,
- NDT results are strongly dependent on the ability of the technician to apply the NDT process, and
- many NDT results cannot be cross-checked without conducting another test.

NDT is typically the last step in the construction/fabrication process. Rejectable discontinuities may go undiscovered until failure occurs. This results in a high level of responsibility to ensure testing is performed correctly and all relevant indications are located and addressed.

The Preamble to the NDT Level III *Code of Ethics* states, "In order to safeguard the life, health, property, and welfare of the public, to maintain integrity and high standards of skills and practices in the profession of nondestructive testing, the following rules of professional conduct shall be binding upon every person issued a certificate by ASNT as a Level III." It then goes on to discuss integrity, responsibility to ASNT, responsibility to the public, public statements, conflict of interest, solicitation of employment, improper conduct, unauthorized practices, and rulings of other jurisdictions. Failure to comply with this code can result in sanctions up to and including revocation of certification.

The primary method of NDT certification in the United States is employer-based. The principal document for employer-based certification schemes used to ensure that NDT personnel are properly qualified and certified is the *written practice* in SNT-TC-1A and the written *certification procedure* in CP-189. A written practice (or procedure) is nothing more than the written certification procedure that describes the processes that the employer will use to train, qualify, and certify their NDT personnel. If certification is to be in accordance with Recommended Practice No. SNT-TC-1A, the written practice must also document any modifications to the detailed recommendations found in SNT-TC-1A and give the rationale for such modifications in an appendix to that practice. Once SNT-TC-1A *guidelines* are adopted in a written practice, they become the certification *requirements* for that company.

If certification is to be in accordance with a standard such as ANSI/ASNT CP-189, then the written certification procedure (practice) must describe how each of the requirements will be administered without modification. In both cases, the written practice must be approved by the employer's NDT Level III.

The written practice must also document that qualification examinations will be administered that are suitable to demonstrate that the individual has acquired sufficient knowledge and capabilities for the applicable qualification level, and that certification will be issued to document successful qualification. The written practice should also describe how experience will be documented and which qualified personnel will be responsible for supervising that experience.

In many nations, a central approach to certification is being adopted using internationally recognized certification standards such as ISO 9712. The certifying body may be a government, or an accredited organization recognized by the government. While central certification systems exist in the US, such as those for the ASNT Central Certification Program (ACCP), the majority of US NDT certification programs today are employer-based and in accordance with ASNT's Recommended Practice No. SNT-TC-1A, with a smaller number of companies choosing to use the standard ANSI/ASNT CP-189.

For training outlines, both SNT-TC-1A and ANSI/ASNT CP-189 refer to the ASNT body of knowledge that is found in ANSI/ASNT CP-105. For this reason, it is important to understand these documents and how they are applied in a few of the widely used specifications that require NDT personnel certification.

Training Outlines

It is important to understand ANSI/ASNT CP-105 and ISO/TS 25107 and how they are applied.

ANSI/ASNT CP-105

ANSI/ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* was initially processed and approved for submittal to the American National Standards Institute (ANSI) by the ASNT Standards Council (StC). ANSI/ASNT CP-105 is a US standard that establishes the minimum topical outline requirements for the qualification of NDT personnel. It details the minimum training course content for NDT personnel. These topical outlines are progressive, meaning consideration as NDT Level I is based on satisfactory completion of the Level I training course and consideration as NDT Level II is based on satisfactory completion of both Level I and Level II training courses. Topics in the outlines may be deleted or expanded to meet the employer's specific applications or for limited certification, unless stated otherwise by the applicable codes, standards, regulatory authority, certification procedure, and/or written practice. Topical outlines included are:

- acoustic emission testing (AE)
- electromagnetic testing (ET)
  - alternating current field measurement
  - eddy current testing
  - remote field testing
- ground penetrating radar (GPR)
- guided wave testing (GW)
- laser methods testing (LM)
  - profilometry
  - holography/shearography
- leak testing (LT)
  - bubble leak testing
  - halogen diode leak testing
  - mass spectrometer leak testing
  - pressure change leak testing
- liquid penetrant testing (PT)
- magnetic flux leakage testing (MFL)
- magnetic particle testing (MT)
- microwave technology testing (MW)
- neutron radiographic testing (NR)
- radiographic testing (RT)
  - computed radiography
  - computed tomography
  - digital radiography
  - radiography
  - limited radiographic film interpretation
- thermal/infrared testing (IR)
- ultrasonic testing (UT)
  - full matrix capture
  - phased array ultrasonic testing
  - time of flight diffraction
  - digital thickness measurement limited
  - A-scan thickness measurement limited
- vibration analysis (VA)
- visual testing (VT)
- basic
- predictive maintenance (PdM)

Note: Appendix A addresses radiographic safety.

Training should be suited to the employer's needs; it should include the actual processes and products, and it should relate specifically to the appropriate level of certification. The recommended training course outlines in ANSI/ASNT CP-105 can be used as a guide; however, some expertise will be needed to adapt them to special cases. CP-105 topical outlines should only be modified by companies whose procedures and practices are limited to specific elements of a method or exceed those topics listed. (Example: ABC Manufacturing regularly performs VT; however, the procedures do not utilize rigid borescopes but do utilize custom visual inspection gauges. Therefore, ABC Manufacturing removes the topical outline for rigid borescopes and incorporates topics on the specific visual gauges from its training curriculum.)

ISO/TS 25107

ISO/TS 25107 (an international document) was prepared by ISO/Technical Committee 135, Non-destructive testing, Subcommittee 7, Personnel Qualification. This technical specification was recommended by ISO/TC 135 to the worldwide NDT community stating it contains the minimum technical knowledge to be required of NDT personnel. These recommendations provide means for evaluating and documenting the competence of personnel whose duties demand the appropriate theoretical and practical knowledge.

To streamline and harmonize the training and certification of NDT personnel, ISO/TC 135 developed guidelines for training syllabuses (ISO/TS 25107). This document is intended to serve those involved in training and to be useful in achieving a uniform level of training material and consequently in the competence of personnel. The 2019 edition of ISO/TS 25107 addresses the following NDT methods and levels:

- radiographic testing (RT) – Levels 1, 2, and 3
- ultrasonic testing (UT) – Levels, 1, 2, and 3
- eddy current testing (ET) – Levels 1, 2, and 3
- penetrant testing (PT) – Levels 1, 2, and 3
- magnetic particle testing (MT) – Levels 1, 2, and 3
- leak testing (LT) – Levels 1, 2, and 3
- acoustic emission testing (AT) – Levels 1, 2, and 3
- visual testing (VT) – Levels 1, 2, and 3
- thermographic testing (TT) – Levels 1, 2, and 3
- strain gauge testing (ST) – Levels 1, 2, and 3

There are also provisions for developing techniques which provide recommendations on training pertaining to ultrasonic time of flight diffraction (UT-TOFD), ultrasonic phased array (UT-PA), and MFL. "Annex A (informative) Alternative training hours for advanced radiographic techniques" addresses the RT: radiographic testing method, RT-F: film technique, RT-D: digital technique (film replacement), and RT-S: radioscopic technique.

Note: ISO 9712 uses Level 1, 2, and 3 identifiers in lieu of Roman numeral designations.

Certification Schemes

There are two types of certification schemes in the NDT industry: employer-based and central certification.

Employer-Based versus Central Certification

Employer-based certification is a system, when permitted by code, standard, and/or regulatory requirements, that allows the employer to define how NDT personnel are qualified and certified to perform one or more NDT methods. This includes the employer's right to define basic attributes for qualification (for example, educational requirements, training, experience, etc.) and to develop, administer, and grade their own qualification examinations. Examples of typical employer-based certification schemes include the following:

- SNT-TC-1A is the world's most widely used and accepted employer-based certification. It is recognized by most national and many international codes and standards. As a guideline, it allows employers to modify their programs to fit their needs. But any modifications should be addressed/justified in their qualification/certification procedures.
- ANSI/ASNT CP-189 is also an employer-based certification. Unlike SNT-TC-1A, CP-189 is a US standard. As a standard, it cannot be modified.
- NAS 410 is an employer-based certification specific to the aerospace industry. NAS 410 is a standard and cannot be modified**.**

The recommended training for each method is typically addressed in a referenced document. Example: For the previous two ASNT schemes, ANSI/ASNT CP-105 is referenced. These outlines were developed by ASNT's Technical and Education (T&E) Council. The council is made up of method-specific subcommittees made up of method-specific subject matter experts. One responsibility of each subcommittee is to develop topical outlines to be incorporated/updated in the next edition of ANSI/ASNT CP-105. Once the outlines are complete, the method subcommittee sends the recommended outline to ASNT's Standards Council (StC). The StC compiles the various method outlines into one document which will become the next edition of CP-105. Using the ANSI consensus process, the StC submits the draft of CP-105 for review and comment by interested parties. All comments received from this process are evaluated and dispositioned. Once all comments have been dispositioned, the final draft of CP-105 is submitted to the StC for approval.

NOTE

Comments received in the consensus process are submitted to the applicable T&E method subcommittee for resolution.

In an employer-based system, a person's certification is terminated when the individual leaves that company.

The employer is always responsible for the employee's performance.

A problem with employer-based certification is that the implementation of the qualification process varies from employer to employer. Since it is applicable to the employer only, there is no outside evaluation of its content, difficulty level, and/or an evaluation of performance issues that leads to retraining/retesting/recertifying. Finally, there is no way to evaluate the effectiveness and quality of written or practical examinations. Another issue is the expense associated with training and practical examination samples.

Central certification is a system where the qualification examinations are administered by a third-party certification body. Certificates issued by a certification body indicate that the certificate holder has fulfilled the minimum qualification examination requirements for the applicable certification system. The main advantage of central certification is that it provides standardized, third-party examinations developed and administered by an independent certification body.

Central certification schemes for NDT personnel include, but are not limited to:

- ASNT Central Certification Program
- ISO 9712: *Non-destructive testing – Qualification and certification of NDT Personnel*
- ANDE-1 – *ASME Nondestructive Examination and Quality Control Central Qualification and Certification Program*
- American Petroleum Institute (API) – NDT and Visual Certifications
- American Welding Society (AWS) – NDT and Visual Certifications
- International Code Council (ICC) – Bolting and Structural Steel

Certificates issued by a certification body are transferable and do not automatically expire when NDT personnel change employers.

Central certification is a step forward for standardization; however, true standardization has not yet been achieved. Even where ISO 9712 has been adopted, there still remains variation in quality of the compliant ISO 9712 program. Independent oversight is needed to verify program compliance and evaluate the quality of certified NDT personnel performance. Convergence of certifications would allow better international transferability of personnel.

Employer-Based Schemes

This section explains examples of employer-based certification schemes including SNT-TC-1A, CP-189, ASNT ILI-PQ, and NAS 410.

RECOMMENDED PRACTICE NO. SNT-TC-1A

SNT-TC-1A is a recommended practice establishing the general framework for an employer-based qualification and certification program. The document provides educational, experience, and training recommendations for the different NDT test methods. This recommended practice is not intended to be used as a strict specification. It is recognized, however, that contracts require programs that meet the intent of this document. For such contracts, purchaser and supplier must agree upon the acceptability of an employer's program.

This document addresses the qualification and certification requirements for the following methods:

- acoustic emission testing
- electromagnetic testing
- ground penetrating radar
- guided wave testing
- laser methods testing
- leak testing
- liquid penetrant testing
- magnetic flux leakage testing
- magnetic particle testing
- microwave technology testing
- neutron radiographic testing
- radiographic testing
- thermal/infrared testing
- ultrasonic testing
- vibration analysis
- visual testing

ANSI/ASNT CP-189 – AMERICAN NATIONAL STANDARD

ASNT has undertaken the preparation and publication of ANSI/ASNT CP-189: *ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel* to specify the procedures, essential factors, and minimum requirements for qualifying and certifying NDT personnel. An essential element in the effectiveness of NDT is the qualification of the personnel who are responsible for and who perform NDT. Formal training and actual experience are important and necessary elements in acquiring the skills necessary to effectively perform nondestructive tests. ANSI/ASNT CP-189 applies to personnel whose specific tasks or jobs require appropriate knowledge of the technical principles underlying NDT methods for which they have responsibilities within the scope of their employment. These specific tasks or jobs include, but are not limited to, performing, specifying, reviewing, monitoring, supervising, and evaluating NDT work.

CP-189 was processed and approved for submittal to ANSI by the ASNT StC.

This document addresses the qualification and certification requirements for the following methods:

- acoustic emission testing
- electromagnetic testing
- leak testing
- liquid penetrant testing
- magnetic flux leakage testing
- magnetic particle testing
- neutron radiographic testing
- radiographic testing
- thermal/infrared testing
- ultrasonic testing
- vibration analysis
- visual testing

ANSI/ASNT ILI-PQ

ANSI/ASNT ILI-PQ: *ASNT In-Line Inspection Personnel Qualification and Certification* has been developed by ASNT to establish minimum requirements for the qualification and certification of in-line inspection (ILI) personnel whose jobs require specific knowledge of the technical principles of ILI technologies, operations, regulatory requirements, and industry standards as applicable to pipeline systems.

This document addresses the qualification and certification requirements of Levels I, II, and III tool operators and data analysts for the following ILI technologies and positions:

- geometry technology
- axial magnetic flux technology
- transverse magnetic field technology
- ultrasonic compression wave technology
- ultrasonic shear wave technology
- EMAT technology
- mapping technology

Qualification and certification of personnel shall be the responsibility of the employer.

NAS 410 – NATIONAL AEROSPACE STANDARDS

The National Aerospace Standards (NAS) are voluntary standards developed by the aerospace industry. NAS 410: *NAS Certification & Qualification of Nondestructive Test Personnel*, better known as NAS 410, is the standard in which all NDT personnel are qualified to in the aerospace industry.

NAS 410 began as MIL-STD-410 for the certification of active-duty military personnel performing NDT. Department of Defense specifications and standards were replaced by industry specifications in 1994.

This standard establishes the minimum requirements for the qualification and certification of personnel performing NDT, NDI, or NDE in the aerospace manufacturing, service, maintenance, and overhaul industries.

NAS 410 uses the same general concepts as SNT-TC-1A but unlike SNT-TC-1A, NAS 410 is a not a recommended practice—and must be fully complied with.

Following is a brief outline of the more pronounced differences between NAS 410 and SNT-TC-1A to assist with the understanding of this document.

- **Methods:** NAS 410 only addresses liquid penetrant testing, magnetic particle testing, thermographic testing, eddy current testing, ultrasonic testing, and radiographic testing. Organizations may certify technicians in other methods; however, the responsibility falls to the responsible NDT Level III.
- **Certification Levels:** Four basic levels of certification exist: NDT Level I Limited, NDT Level I, NDT Level II, and NDT Level III.
- **Training Hours:** Training hours differ slightly for some methods compared to SNT-TC-1A. For example, NAS 410 requires 16 hours for Level I and 16 hours for Level II in PT compared to 4 and 8 hours in SNT-TC-1A. For UT, the hours are the same—40 for Level I and 40 for Level II; however, advanced NDT techniques such as phased array ultrasonic testing and TOFD are not specifically addressed. Level III training hours are not required beyond that of Level II unless it is in a method other than those addressed. For methods not addressed or emerging methods, a minimum number of training hours is required for Level III.
- **Experience:** Hours of experience for NAS 410 are not separated into minimum hours in method versus total hours in NDT as in SNT-TC-1A. Experience is simplified into minimum hours required for Level I and Level II. For example, in PT Level I, SNT-TC-1A outlines 70 hours in the method and 130 total hours of NDT experience. NAS 410 simply requires 130 hours of experience. For Level III, NAS 410 and SNT-TC-1A experience hours are the same.
- **Examinations:** NAS 410 requires a General, Specific, and Practical examination for each level certified. One unique element for practical examinations is that if the candidate is only required to interpret the results and not perform the process of generating the image, the test samples may be images, such as radiographs or other resultant test data.
- **National Aerospace NDT Board (NANDTB):** One item within NAS 410 that has generated a lot of confusion is the NANDTB. This is defined as an independent aerospace organization representing a nation's aerospace industry that is chartered by the participating prime contractors and recognized by the nation's regulatory agencies to provide or support NDT qualification and/or examination services. NANDTBs are quite common outside of the United States but within the US, NANDTBs are not used and therefore not applicable for companies within the US.

CENTRAL CERTIFICATION SCHEMES

This section explains examples of central certification schemes including ISO-9712, ASNT NDT Level II, ASNT NDT Level III, ACCP, ASNT IRRSP, ASNT ISQ, API – NDT and Visual Certification, AWS – NDT and Visual Certification, ICC – Bolting and Structural Steel, and the Canadian General Standards Board NDT Certification.

ISO 9712

The International Organization for Standardization (ISO) is a worldwide federation of national standards bodies (ISO member bodies). ISO 9712: *Non-Destructive Testing – Qualification and Certification of NDT Personnel* has worldwide use—many countries have their own scheme based on ISO 9712.

ISO 9712 is an international document prepared by Technical Committee ISO/TC 135, Non-destructive testing, Subcommittee SC7, Personnel Qualification. This standard was presented by ISO/TC 135 to the worldwide NDT community with their recommendations for the minimum qualification and certification requirements to be required of NDT personnel. This international standard provides a means of evaluating and documenting the competence of personnel whose duties require the appropriate theoretical and practical knowledge of the nondestructive tests they perform, specify, supervise, monitor, or evaluate. To implement ISO 9712, each country must formally adopt it. It can be adopted as written without modification within the country's national standards process, or the country can adopt by the development of a national standard that addresses each ISO 9712 requirement and identify where/why modifications were made. Acceptable modifications are those that do not conflict with the country's applicable laws of the land.

When certification of NDT personnel is required in product standards, regulations, codes, or specifications, the employer may choose or may be required to certify their personnel in accordance with ISO 9712. When ISO 9712 is required, the following methods are addressed:

- acoustic emission testing
- eddy current testing
- leak testing (hydraulic pressure tests excluded)
- magnetic testing
- penetrant testing
- radiographic testing
- strain gauge testing
- thermographic testing
- ultrasonic testing
- visual testing (direct unaided visual tests and visual tests carried out during the application of another NDT method are excluded)

ASNT NDT Level II

The ASNT NDT Level II certification program was developed to provide standardized Level II written examinations that employers may use to satisfy the General and Specific examination guidelines of Paragraphs 8.3 and 8.4, respectively, of SNT-TC-1A for MT, PT, RT, UT, and VT.

- **Multipart Examination:** Successful ASNT NDT Level II certification candidates must complete the General examination and at least one Specific examination.
- **General Examinations:** The 50-question multiple-choice Level II General examinations cover the fundamentals, principles, and theory found in the applicable Level II topical outlines found in ANSI/ASNT CP-105. The General examination is the same for all sectors for any one test method.

- **Specific Examinations:** ASNT currently offers Specific examinations for the general inspection and pressure equipment sectors. These examinations consist of 40 multiple-choice questions based on an NDT procedure covering the equipment, operating processes, and NDT techniques commonly used in the applicable industry sector.
- **Practical Examinations:** This is the employer's responsibility.

ASNT NDT Level III

The ASNT NDT Level III program provides third-party certification for NDT personnel whose specific jobs require knowledge of the technical principles underlying the nondestructive tests they perform, witness, monitor, or evaluate. The program provides a system for ASNT NDT Level III certification in accordance with SNT-TC-1A. An ASNT NDT Level III certificate holder shall have:

- the skills and knowledge to establish techniques; to interpret codes, standards, and specifications; and to prepare or approve procedures and instructions.
- general familiarity with other NDT methods; shall be capable of conducting or directing the training and examination of testing personnel in the methods for which the ASNT Level III is qualified; and shall have knowledge of materials, fabrication, and product technology in order to establish techniques and to assist in establishing acceptance criteria when none are otherwise available.

Exams and methods available are:

- basic
- acoustic emission testing (AE)
- electromagnetic testing (ET)
- liquid penetrant testing (PT)
- magnetic flux leakage (MFL)
- magnetic particle testing (MT)
- radiographic testing (RT)
- thermal/infrared testing (IR)
- ultrasonic testing (UT)
- visual testing (VT)

Note: ASNT NDT Level IIIs are required to pass both the Basic examination and a method examination to receive the ASNT NDT Level III certificate.

ACCP

The purpose of the ACCP Level II and III certifications is to cover Practical and Procedure Preparation examinations in five NDT methods (MT, PT, RT, UT, and VT). The Level III also includes the Basic examination. The ACCP provides the NDT industry with personnel who have achieved a high standard of NDT qualifications by examination and independent, transportable NDT certifications.

ACCP's approach to NDT personnel qualification is unique in the industry. Using the T&E Council method committees in conjunction with the StC ANSI consensus process, ANSI/ASNT CP-105 was developed and approved as a US standard. The Certification Management Committee (CMC) uses CP-105 and a detailed job task analysis as the foundation for the ACCP examination process. Using psychometric principles, the CMC develops the ACCP examinations ensuring each examination is fair and of equal difficulty.

The ACCP program is in the process of being renamed ASNT-9712 and is being revised to meet the ISO 9712-2021 edition.

ASNT IRRSP

The Industrial Radiography Radiation Safety Personnel (IRRSP) program provides third-party radiation safety certification that meets the certification requirements of both Title 10 Part 34 of the *Code of Federal Regulations* (10CFR34), and the Suggested State Regulations for Control of Radiation (SSRCR) – Part A, *General Provisions*; Part E, *Radiation Safety Requirements for Industrial Radiographic Operations*; Part H, *Radiation Safety Requirements for Non-Healing Arts Radiation Generating Devices*; Part T, *Transportation of Radioactive Material*.

Three IRRSP certification examinations are available:

- radioactive materials (RAM) for personnel using gamma sources only;
- X-ray for personnel using X-ray machines only; and
- combination (combo) for personnel working with both RAM and X-ray devices.

Acceptance of the IRRSP radiation safety certification within a state is solely at the discretion of the state in question. However, states that do recognize *approved certifying entities* will accept the IRRSP certifications because ASNT is an approved Certifying Entity.

ASNT Industry Sector Qualification Program (ISQ)

The Industry Sector Qualification - Oil & Gas (ISQ-O&G) program is an advanced practical qualification exam. The purpose of the ISQ program is to provide the oil and gas industry with NDT personnel who have demonstrated competency in practical application of a specific technique through hands-on performance-demonstration qualification examinations.

The ISQ program

- standardizes performance-demonstration initiative testing of NDT technicians;
- alleviates the burden of owner/operators from providing their own NDT performance-demonstration program and provides global availability;
- establishes an industry-recognized program for the oil and gas sector while minimizing the impact of cost and redundant testing of NDT technicians; and
- currently offers UT thickness, which emphasizes the candidate's ability to distinguish wall loss from laminations and accurately measure remaining wall thickness, potentially saving hundreds of thousands of dollars in repairs or loss of production
- conducts ISQ exams through ASNT Authorized Exam Centers

Note: While ASNT has implemented this qualification with the support of major oil and gas companies, the program will continue to strive to meet the standards of all participating owner/operator representatives, and to achieve universal industry adoption.

API – NDT and Visual Certification Schemes

API has several central certification programs specific to the petroleum industry. API's Individual Certification Programs have provided the petroleum and petrochemical industries with an independent and unbiased way to evaluate the knowledge and experience of technical and inspection personnel. These certification programs are based on the industry-developed standards that are recognized and used with confidence worldwide.

The majority of API's ICPs consist of visual inspection certifications based on specific elements within the petroleum industry. Some of the most popular visual certification schemes are:

- API 510 – Pressure Vessel Inspector
- API 570 – Piping Inspector
- API 653 – Aboveground Storage Tank Inspector
- API 1169 – Pipeline Construction Inspector

The experience and education requirements vary for each API certification, but they all follow the same general scheme. Once the experience and education requirements are met, the inspector is permitted to take the examination.

In addition to visual-based inspector programs, API also offers central certifications for UT specific to the petroleum industry. They are:

- API QUTE – Qualification of Ultrasonic Testing Examiners (Detection)
- API QUTE-TM – Qualification of Ultrasonic Testing Examiners (Thickness Measurement)
- API QUSE – Qualification of Ultrasonic Testing Examiners (Sizing)
- API QUSE-PA – Qualification of Ultrasonic Testing Examiners (Crack Sizing)
- API QUPA – Qualification of Ultrasonic Testing Examiners (Phased Array)

AWS – NDT and Visual Certification Schemes

AWS provides several central certification schemes for NDT, the most popular of which is the Certified Welding Inspector (CWI) and its variants. The CWI program is a visual inspector certification specific to welding inspection in the structural industry. There are three levels: Certified Associate Welding Inspector (CAWI), CWI, and Senior Certified Welding Inspector (SCWI).

The eligibility requirements for the CAWI and CWI levels are based on a certain number of years' experience in the welding or NDT industries. The number of years vary depending on the base education level. For example, a high school graduate would need two years of industry experience to qualify to take the CAWI. For the CWI, a minimum of five years' industry experience is required if a high school graduate. The SCWI is a bit different in that it is required to have been certified as a CWI for a minimum of six years and obtained a minimum of 15 years' industry experience.

The CAWI and CWI examination are identical, the difference being the CAWI requires a minimum score of 60% for each part and an overall minimum score of 60%. The CWI requires a minimum score of 72% for each part and an overall minimum score of 72%.

The exams consist of three parts, Part A – Fundamentals, Part B – Practical, and Part C – Code book.

- Part A is a closed-book exam consisting of a minimum of 150 multiple-choice questions in a 2-hour time limit.
- Part B is an open-book exam consisting of a minimum of 40 multiple-choice questions in a 2-hour time limit.
- Part C is an open-book exam consisting of a minimum of 46 multiple-choice questions in a 2-hour time limit.

The SCWI exam consists of two parts—A (Technical Fundamentals) and B (Administrative Fundamentals)—consisting of 200 multiple-choice questions in an open-book format with a 4-hour time limit.

In addition to the visual welding inspector certifications AWS offers *endorsements* for currently certified CWIs and SCWIs in MT and PT.

The MT and PT endorsements do not follow the typical Level I, II, III format. The only prerequisite for these endorsements is a current CWI or SCWI.

Additionally, the MT and PT endorsements are limited to visual dry powder AC yoke technique for MT and Type II Method (C) Form (e) PT. The examinations consist of a closed-book 40-question fundamentals exam, a 20-question open-book Specific exam (based on the AWS PT or MT *Book of Specifications*) and a Practical exam consisting of two test specimens.

AWS also offers the Certified Radiographic Interpreter (CRI) certification. The prerequisites for this stand-alone certification are a minimum one year of experience as a company-certified or nationally certified individual in radiographic interpretation, or the performance of radiographic interpretation under direct supervision. Additionally, a CRI is required to be a high school graduate and have completed 40 hours of organized training in radiographic interpretation/examination.

The examination consists of three parts:

- Part A – A multiple-choice examination of the fundamentals of radiographic inspection.
- Part B – A multiple-choice practical examination consisting of interpreting a minimum of 10 radiographs to a code, specification, or other standard.
- Part C – A multiple-choice, open-book examination covering the contents of specific codes that pertain to radiographic quality and film interpretation.

The CRI examination is 6 hours in total, Parts A and C are administered together in a single 3-hour session. Part B is administered in a single 3-hour session.

ICC – Bolting and Structural Steel

The ICC offers several central certifications related to building, plumbing, electrical inspection, and more. Additionally, the ICC offers what it refers to as *Special Inspector* certifications based on visual inspections of vertical structures. Currently, there are eight different Special Inspector certifications: Reinforced

Concrete, Structural Masonry, Prestressed Concrete, Spray-applied Fireproofing, Tall Mass Timber Buildings, Foundation Soils, Structural Steel/Bolting, and Structural Welding.

Of these, the Structural Steel/Bolting and Structural Welding Special Inspector certifications are the most closely aligned with the NDT industry. They are both visual inspection certifications restricted to the structural industry, specifically vertical steel structures governed by the International Building Code.

Each Special Inspector exam follows the same format consisting of three parts.

- Special Inspector General Requirements – 25 multiple-choice questions, open-book, 1-hour time limit.
- Codes Module – 60 multiple-choice questions, open-book, 2-hour time limit.
- Plans Module – 30 multiple-choice questions, open-book, 1.5-hour time limit.

There are no prerequisites for the Structural Steel/Bolting exam. The prerequisite for the Structural Welding exam is to be a current Structural Steel/Bolting Special Inspector.

Canadian General Standards Board NDT Certification (CGSB)

In Canada, NDT certification is regulated and governed by the Canadian Government through the CGSB.

The CGSB is part of Public Works and Government Services Canada. It develops standards and provides conformity assessment services to government entities and private-sector companies.

In addition to developing product standards, CGSB's other primary function is to set and maintain standards for certifying individuals in NDT roles.

Accredited by the Standards Council of Canada, the CGSB sets certain criteria for NDT skills such as ultrasonic testing, radiographic testing, magnetic particle testing, liquid penetrant testing, and eddy current testing.

Natural Resources Canada (NRCan) manages the national program in Canada that certifies people who perform NDT.

NRCan's National Non-Destructive Testing Certification Body (NDTCB) provides:

- CGSB certification for nondestructive testing.
- Portable tube-based X-ray fluorescence analyzer operator certification.
- The written examination for the Canadian Nuclear Safety Commission's exposure device operator certification (NRCan NDTCB is the qualifying body for this examination).
- Training and examinations for CBSB certification must be taken through an organization recognized by NRCan.

Codes and Standards

This section discusses how SNT-TC-1A and CP-189 can be applied to some of the more frequently used industrial specifications. Each specification will be discussed separately.

American Society of Mechanical Engineers – *ASME Boiler & Pressure Vessel Code*

The *ASME Boiler & Pressure Vessel Code* (Code–2019 Edition) has many sections, each with different requirements. The personnel qualification and certification (PQ&C) requirements of different sections of the Code can vary. Some sections demand little while others are quite detailed. For example, *ASME BPVC,* Section V: *Nondestructive Examination*, Article 1, *General Requirements*, T-120 identifies the 2016 edition of SNT-TC-1A (as amended by *Mandatory Appendix III*) or ANSI/ASNT CP-189 (as amended by *Mandatory Appendix IV*). Section V requirements are used unless addressed directly by other Code sections. For emerging technologies (for example, computed radiography [CR], digital radiography [DR], phased array ultrasonic testing [PAUT] ultrasonic time of flight diffraction [TOFD], or ultrasonic full matrix capture [FMC]), Section V, *Mandatory Appendix II* addresses training, experience, and examination requirements.

When performing NDE to *ASME BPVC,* Section III Code (Nuclear Construction), SNT-TC-1A (Table NCA-7100-2 identifies 2006 and 2011 editions) certification is required. However, *ASME BPVC,* Section III takes exception to the use of ASNT's ACCP to fulfill any part of the qualification process. Keep in mind that other versions of the Code may differ from the requirements reviewed here.

On the other hand, *ASME BPVC,* Section XI: *Rules for Inservice Inspection of Nuclear Power Plant Components* has considerable PQ&C demands. Section XI, IWA-1600, specifies that the PQ&C written practice (*written practice* is the term used in SNT-TC-1A and *written procedure* is used in CP-189) shall comply with ANSI/ASNT CP-189 or the ACCP and the employer's written practice may be used, with some exceptions. The exceptions are:

- All Level I, II, and III personnel will be qualified and certified, or recertified, by examination only. Recertification will be completed every five years (IWA-2314), as recommended by CP-189 and required by the ACCP and the ASNT NDT Level III program. A composite score of all examinations must be 80% or greater using simple averaging to form the overall score. When an outside agency administers the examinations and only a pass or fail grade is given, the employer will assign a grade of 80% for a pass grade.
- Level II requirements may be partially met by an ACCP certificate with current endorsements obtained by examination and may be used to satisfy the General and Practical examination requirements.
- Level IIIs can only be qualified by written and demonstration examinations including the Basic, Method, Specific, and Practical. Also, a demonstration examination as is required for a Level II Practical must be given. In addition, this area takes exception to the requirement in CP-189 for an NDT Level III to possess a currently valid ASNT NDT or PdM Level III certificate or ACCP Professional Level III certificate. This means that the employer may use other outside agencies as defined in Appendix VII for the required examinations including the

Basic and Method. If an ASNT NDT Level III certification is used to fulfill the Basic and Method examinations required, proof must be given that examinations were taken and passed. The ACCP Professional Level III certification may also be used to fulfill the Basic, Method, Practical, and demonstration examination requirements with proof that examinations were taken and passed.

Level III recertification requires completion of all of the examinations required for initial certification, but may be completed with only a Method and Specific examination if all of the following are met:

- The candidate was previously certified or recertified using all the written and demonstration examinations.
- The candidate is not being recertified due to interrupted service as defined in the employer's written practice.
- The candidate is not being certified by a new employer.

Although new ASNT certifications can be obtained only by examination, recertifications can be obtained either by examination or by documentation of continued NDT activities plus increased knowledge through education or training activities. ASNT certificates do not identify which recertification method was used. However, the ASNT letter advising that the candidate has been recertified does state whether or not an examination was passed and may be used as the necessary proof.

Per IWA-2370, a Level III candidate will have a minimum of a high school diploma or equivalent and will meet one of the following requirements:

- a four-year degree in engineering or science from an accredited college or university plus one year of experience in NDT in an assignment comparable to that of a Level II in the examination method,
- completion with a passing grade of at least the equivalent of two full years of engineering or science study at a university, college, or technical school, plus two years of experience in NDT in an assignment comparable to that of a Level II in the examination method, or
- four years of experience in NDT in an assignment comparable to that of a Level II in the examination method.

Per IWA-2321, all vision tests will be administered annually and meet the following requirements:

1. Near-distance acuity of 20/25 or greater Snellen fraction with at least one eye will be required. The test chart used shall have a measurement of one of the near-distance lower case characters without ascender or descender (a, c, e, and o, for example) measured prior to initial use with an optical comparator (10× or greater). This measurement will be documented and traceable to the test chart. Section XI also provides Table IWA-2322-1 with test distances and maximum lower case character height to evaluate test charts, including Jaeger charts and their equivalents.
2. For visual examinations, VT-2 or VT-3 personnel will demonstrate far distance acuity of 20/30 or greater Snellen fraction or equivalent with at least one eye.
3. Demonstrate the capability to distinguish colors applicable to the NDT methods for which certified and to differentiate contrast between these colors.

For NDT methods listed in CP-189:

- Qualifications are based on the methods, techniques, procedures, and equipment used for the NDT required.
- Training, qualification, and certification of UT personnel shall also comply with the requirements of Appendix VII.
- Training, qualification, and certification of VT personnel shall also comply with the requirements of Appendix VI and the training and experience hours required by CP-189 will cover the VT-1, VT-2, and VT-3 techniques combined. This means that if only one VT technique is certified, it will be treated as a limited certification.

NDT methods not addressed in CP-189 will be qualified as defined in CP-189 or the ACCP and the employer's written practice.

Note: This process is not defined in CP-189 as referenced in this edition of Section XI and would need to be defined in the employer's written practice.

Section XI allows limited certifications, and these limitations must be described in the employer's written practice. Topics not relevant to the limited certification may be deleted from the training requirements in CP-189, Appendix VI, or Appendix VII training outlines, and a corresponding reduction in training hours, examination content, and number of examination questions may also be acceptable in this section. There is an alternative qualification process for VT-2 and VT-3 personnel available in Section XI. To use it the employer must describe specifically the protocol for this alternative process in the written practice.

A difference that may have a major impact on some users is that Level I personnel may not independently evaluate results, contrary to Paragraph 3.5 of CP-189. Thus, for Section XI work, it may be necessary to train and qualify additional Level II personnel.

The requirements for an NDT Instructor in Section XI require the individual to satisfy the Level III Basic and Method examination requirements, and meet one of the following:

- maintain a current teacher or vocational instruction certificate issued by a state, municipal, provincial, or federal authority, or
- complete a minimum of 40 hours of instruction in training and teaching techniques.

API Standard 1104, *Welding Pipelines and Related Facilities,* 21st Edition

This standard covers the required NDT for pipeline welding and specifies that NDT Level I, II, or III personnel shall be qualified in accordance with SNT-TC-1A, ACCP or any other recognized national certification program, which could include CP-189. This document gives acceptance standards for radiographic testing, magnetic particle testing, liquid penetrant testing, and ultrasonic testing techniques and it allows, but does not specify, other types of NDT. There is ambiguity about the duties of a Level I, but the sole limitation is that only a Level II or Level III may interpret test results. Requalification is required every three years for Level I and Level II, and every five years for Level III.

The vision examination requirements for API 1104 also require that NDT personnel have near-distance acuity in at least one eye capable of reading Jaeger number 1 test chart or equivalent. This requirement is in line with CP-189 and more stringent than the latest requirements in SNT-TC-1A.

For UT only Level II or III certified personnel shall standardize equipment, perform the testing, interpret test results, and evaluate the results per the acceptance/rejection criteria. Only a Level III in UT shall develop the application technique and prepare and approve the test procedure.

For RT only Level II or III radiographers shall interpret radiographic images of production welds.

For MT and PT there is no mention of levels of qualifications and their duties.

AWS D1.1/D1.1M 2020: *Structural Welding Code—Steel,* 2020 Edition

This code contains the requirements for the design, inspection, fabrication, and erection of welded steel structures. It requires that NDT personnel be qualified to the current edition of SNT-TC-1A. This document also requires Level III personnel to be certified by ASNT or in accordance with SNT-TC-1A. It also requires that only Level IIs, or Level Is working under a Level II, are permitted to perform NDT. This means a Level III would need to meet the requirements of a Level II including a practical examination if the Level III intends to perform production testing.

For UT personnel specifically, additional qualification requirements are incorporated consisting of specific and practical examinations based on the requirements of D1.1. This is primarily due to the unique testing procedures specific to D1.1 and the use of acceptance criteria tables in lieu of distance amplitude correction curve-based acceptance criteria common in other standards. These tables rely on very specific test parameters that must be followed exactly in order for consistent, accurate examination results.

Though there are no additional qualification requirements for RT personnel like there are for UT, technicians must familiarize themselves with the procedural requirements outlined in this standard. Several elements differ from other industry standards, such as IQI selection and placement, as well as the use of "edge blocks" and different film density requirements. AWS D1.1 also uses some unique acceptance criteria charts and tables, which the technician may not be familiar with.

There is also a requirement for additional experience for the examination of welds using ionizing radiation methods other than RT, such as electronic imaging (including real-time imaging systems) and advanced ultrasonic systems (including multiple probes, multichannel systems, automated inspection, TOFD, and phased array systems). The procedure will be approved by a Level III with a minimum of six months of experience using the same, or similar, equipment and procedures for examination of welds in structural or piping metallic materials. The Level I and II will be certified by the above Level III and have a minimum of three months of experience using the same or similar equipment and procedures for examination of welds in structural or piping metallic materials. Qualification will consist of written and practical examinations to demonstrate the candidate's ability to use the specific equipment and procedures to be used for production examination.

AASHTO/AWS D1.5 *Bridge Welding Code,* 2020 Edition

This American Association of State Highway and Transportation Officials (AASHTO) code covers welding fabrication requirements applicable to welded highway bridges. The code is applicable to both shop and field fabrication of steel bridges and bridge components. The code is to be used in conjunction with the AASHTO *Standard Specifications for Highway Bridges* or the AASHTO *LRFD Bridge Design Specifications*.

All visual inspectors responsible for acceptance or rejection of materials and workmanship shall be qualified as follows:

1. The inspector shall be an AWS CWI qualified and certified in conformance with the provisions of AWS QC1: *Specification for AWS Certification of Welding Inspectors*, or
2. The Inspector shall be qualified by the Canadian Welding Bureau (CWB) to the Level II or Level III requirements of the Canadian Standards Association standard CSA W178.2: *Certification of Welding Inspectors*, or
3. The Inspector shall be an engineer or technician who, by training and experience in metals fabrication, inspection, and testing, is acceptable to the Engineer as an equivalent to (1) or (2).

Personnel performing NDT, other than visual examination, shall be certified in conformance with SNT-TC-1A or an equivalent satisfactory to the engineer. Certification of Level I and Level II individuals shall be performed by the ACCP or a Level III who has been certified by (1) ASNT, or (2) has the education, training, experience, and has successfully passed the written examination prescribed in SNT-TC-1A. Individuals who perform NDT shall be certified as:

- NDT Level II,
- NDT Level I working under the direct supervision of an individual qualified for NDT Level II or NDT Level III, or
- NDT Level III and qualified per SNT-TC-1A as a Level II for NDT performed.

FRACTURE CRITICAL MEMBERS (FCM)

AWS D1.5 has additional qualification requirements for NDT personnel when inspections involve the construction of components deemed *fracture critical*.

Visual inspectors shall be qualified as specified for non-fracture-critical work. Lead inspectors shall have a minimum of three years' experience in steel bridge fabrication inspection. A Lead Inspector shall be defined as the leader of their inspection team at a specific work location, one who assigns other inspectors as necessary, and supervises their work.

NDT technicians shall be certified to Level II or certified to Level III and qualified to perform as a Level II in conformance with SNT-TC-1A. Level II technicians shall be supervised by an individual certified to Level III. Level III individuals shall possess a currently valid ASNT Level III certificate. The engineer may accept alternative qualifications that are deemed equivalent.

Advanced Ultrasonic Examination

Individuals who perform PAUT shall be certified for PAUT per Clause 8 Inspection. Additionally, individuals shall be certified as:

- UT Level II in accordance with 8.1.3.4(1) and have supplemental qualifications for PAUT; or
- UT Level III in accordance with 8.1.3.4(3) and have supplemental qualifications for PAUT. This includes Level III personnel who collect and analyze PAUT data.

Supplemental qualifications for PAUT shall include the following:

- A written test based on the PAUT requirements of this code,
- PAUT examinations of at least two welded joints (for example, butt, T, or corner) that contain real or artificial discontinuities to be examined using a phased array procedure written in accordance with this Annex,
- 320 hours of PAUT work experience.

Individuals meeting the supplemental requirements except the work experience shall have data and reports reviewed and confirmed by a UT Level II with the supplemental PAUT qualifications or UT Level III with the supplemental PAUT qualifications.

CERTIFICATION REQUIREMENTS

Certification of Level II PAUT personnel shall be performed by an NDT UT Level III who both meets the requirements of 8.1.3.4 for PAUT and also has received a minimum of 80 hours of formal training in PAUT.

Summary

NDT is very complex and includes a wide variety of testing methods and techniques. It is a daunting task to ensure technicians are both highly competent and highly capable. It is paramount NDT technicians are proficient at performing complicated tests both accurately and consistently. The qualification and certification schemes for these technicians are equally complex. This guide was developed by NDT experts to help assist those seeking to better navigate the numerous requirements relative to the qualification and certification of personnel performing NDT.

In many nations, a central approach to certification is being adopted using internationally recognized certification standards such as ISO 9712. The certifying body may be a government or an accredited organization recognized by the government. Central certification programs are sometimes called third-party certification. Central certification programs are controlled by a certification body, which confirms that an individual has a base set of skills, knowledge, experience, and competence to perform tasks to published recognized guidelines (ISO 9712 for example). The certification body is a separate entity that ensures an unbiased, fair, and consistent certification program nationally and internationally. Many certification bodies will hold an ISO/IEC 17024 accreditation (ISO/IEC 17024: *Conformity assessment – General requirements for bodies operating certification of persons*) to demonstrate compliance. Certification bodies will include a management committee comprising of experts in various product sectors and industries. The management committee is responsible for developing, reviewing, and maintaining the certification program. This will include approval of examination centers, developing examination questions, and also validating and statistically assessing the examination questions through psychometrics. The use of psychometrics demonstrates to the certification body and ISO 17024 assessors that the examinations are fair, reliable, and valid. Central certifications will follow the NDT individual throughout their NDT career provided they maintain their recertification requirements.

While central certification systems exist in the United States, such as the ACCP, NAVSEA-250-1500-1, API (NDT and Visual Certifications), or AWS (NDT and Visual Certifications), the majority of US NDT certification programs are employer-based. Most employer-based programs are written in accordance with SNT-TC-1A, with a smaller number of companies choosing to use the standard CP-189. The use of SNT-TC-1A is also the foundation of most international employer-based certification programs which is recognized and/or required by international and US codes and standards used abroad.

It should also be noted that most technical documents have evolved since they were created and approved for use. As they evolve, they typically become more complex. Normally each evolution is tracked by *year editions* of adoption and publication. Those performing NDT to national codes and/or standards, should be aware their governing documents are also tracked by specifically identified *editions*.

An example of this evolution is evident in that there are 14 editions of SNT-TC-1A and seven editions of CP-189. One or more of these editions may be encountered, depending on the contract working to and the industry working in. These editions are dated back to 1966 for SNT-TC-1A and 1991 for CP-189. Care must be taken to use the edition that applies to the contract, especially if verbatim compliance with a recommended practice or a standard is required. Since some codes and standards impose additional personnel qualification and certification requirements on the employer, it is of utmost importance to know the requirements and incorporate them into the written practice or certification procedure.

ISBN: 978-1-57117-589-2
Catalog No.: 10178

The American Society for
Nondestructive Testing Inc.